SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4 **Total Marks:40**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:
a) Identify all actors involved in the system.
b) List the primary use cases.
c) Define relationships (include, extend, generalization).
d) Construct a complete Use Case Diagram representing system functionality.

(10 Marks)

2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:
a) Identify classes with attributes and methods.
b) Define relationships (association, aggregation, inheritance).
c) Specify multiplicity between classes.
d) Design a detailed Class Diagram for the system.

(10 Marks)

3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

Tasks:
a) Identify objects involved in the process.
b) Define message flow between objects.
c) Represent activation and lifelines.
d) Construct a Sequence Diagram showing complete interaction flow.

(10 Marks)

4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

Tasks:
a) Identify all activities in the workflow.
b) Add decision nodes for approval/rejection.
c) Include parallel processing where applicable.
d) Design an optimized Activity Diagram.

(10 Marks)

5. State Diagram Problem

A **Vending Machine System** operates through states such as Idle, Product Selected, Payment Received, Product Dispensed, and Transaction Completed.

Tasks:
a) Identify all possible states.
b) Define events triggering transitions.
c) Include initial and final states.
d) Construct a State Transition Diagram representing system behavior.

(10 Marks)

6. Component Diagram Problem

An **Online Learning Platform** consists of modules such as User Management, Course Management, Assessment Module, Content Delivery Module, and Notification Service.

Tasks:
a) Identify major system components.
b) Define interfaces between components.
c) Show dependencies among modules.
d) Construct a Component Diagram representing system architecture.

(10 Marks)

7. Deployment Diagram Problem

A **Smart Traffic Monitoring System** uses roadside sensors to collect vehicle data. The data is transmitted to a central cloud server where traffic authorities monitor congestion through a web dashboard. Alerts are generated during traffic violations and accidents.

Tasks:
a) Identify all hardware nodes involved in the system.
b) Specify software components deployed on each node.
c) Define communication links between nodes.
d) Construct a complete Deployment Diagram representing the system architecture.

(10 Marks)

RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation